Ideation phase

Prosperity Prognosticator: Machine Learning for Startup Success Prediction

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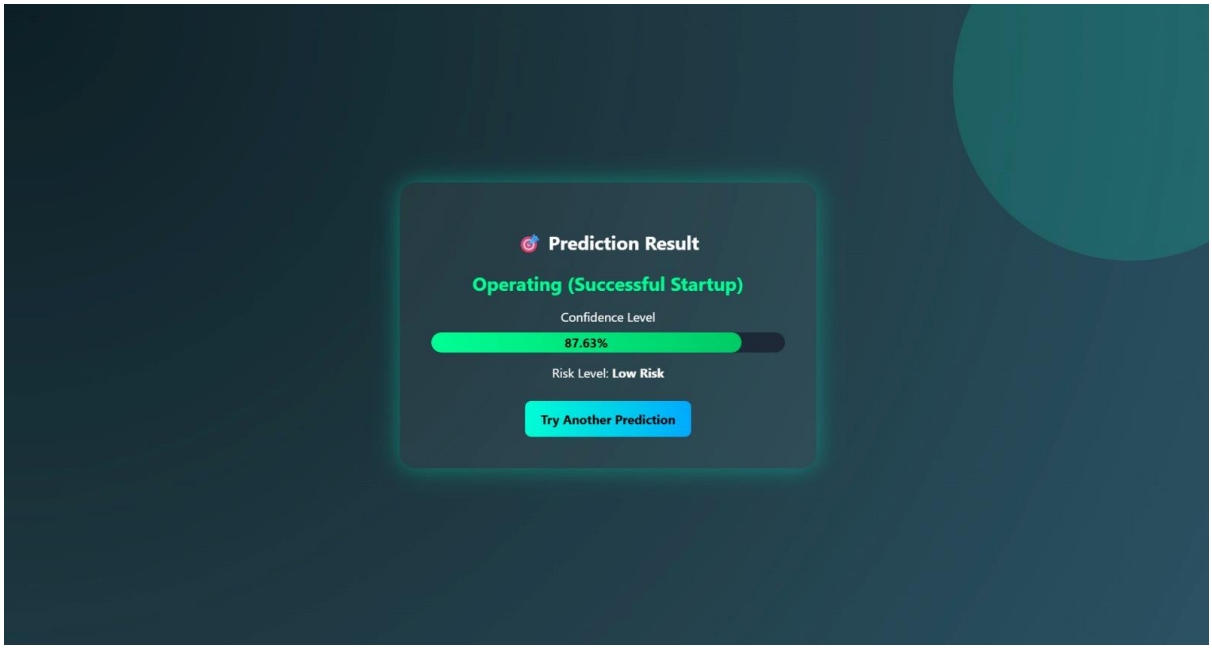
Project Title: Prosperity Prognosticator: Machine Learning for Startup Success Prediction

Brainstorm

STEP-1: Problem Statement:

Despite the surge in global entrepreneurship, the failure rate for new ventures remains high, often exceeding 70–90% within the first few years. Stakeholders—specifically investors, entrepreneurs, and policymakers—face a significant "information asymmetry" problem. They often rely on intuition, fragmented market data, or historical biases to make high-stakes decisions, leading to inefficient resource allocation and lost economic potential.

STEP-2: Output:



STEP-3: Technologies Used:

1. Programming Language

Python: The primary language used for its extensive libraries in data science and machine learning.

2. Data Manipulation & Analysis (The "Data Preprocessing" Stage)

Pandas: Used for loading the .csv file, cleaning missing values, and handling the dataframe.

NumPy: Used for high-performance mathematical operations on the dataset arrays.

Scikit-Learn (Preprocessing): For scaling data (StandardScaler) and encoding categorical variables (LabelEncoding) so the algorithm can understand them.

3. Data Visualization (Understanding the Data)

Matplotlib: The foundational library for creating static plots (histograms, line graphs).

Seaborn: Used for more advanced, "aesthetic" visualizations like heatmaps (to check correlation between startup features) and box plots (to find outliers).

4. Machine Learning Implementation (The "Algorithm" & "Model" Stage)

Scikit-Learn: The core library for implementing algorithms. Based on your project type, you likely used:

Logistic Regression or Random Forest: Popular for binary classification (Success vs. Failure).

Train_Test_Split: To divide the Kaggle data into the "Train Data" and "Test Data" shown in your diagram.

5. Model Evaluation (The "Evaluation" Stage)

Classification Report & Confusion Matrix: To measure precision, recall, and accuracy.

ROC-AUC Curve: To visualize how well the model distinguishes between successful and unsuccessful startups.

6. Deployment & UI (The "User" & "UI" Stage)

Streamlit or Flask: (Commonly used) To create the web interface where a user enters startup "Inputs" and receives a "Prediction."

Pickle/Joblib: To save the trained model so it can be loaded into the UI without retraining